

Appendix B:

Formal definitions

First Fundamental Form:

$$I = \begin{bmatrix} E & F \\ F & G \end{bmatrix} = \begin{bmatrix} 1+h_u^2 & h_u h_v \\ h_u h_v & 1+h_v^2 \end{bmatrix} \text{ with}$$

$$E = \vec{r}_u \cdot \vec{r}_u \quad F = \vec{r}_u \cdot \vec{r}_v \quad G = \vec{r}_v \cdot \vec{r}_v \quad \text{with } \vec{r} = (u, v, h[u, v]) \quad , \quad \vec{r}_u = \frac{\partial \vec{r}}{\partial u} \quad \text{and} \quad \vec{r}_v = \frac{\partial \vec{r}}{\partial v}$$

Second Fundamental Form:

$$II = \begin{bmatrix} L & M \\ M & N \end{bmatrix} = \begin{bmatrix} h_{uu} & h_{uv} \\ h_{uv} & h_{vv} \end{bmatrix} \frac{1}{\sqrt{(1+h_u^2+h_v^2)}} \quad \text{with} \quad g = \|\nabla h \cdot \nabla h^T\| = \sqrt{(1+h_u^2+h_v^2)}$$

$$L = \vec{r}_{uu} \cdot \vec{n} \quad M = \vec{r}_{uv} \cdot \vec{n} \quad N = \vec{r}_{vv} \cdot \vec{n}, \quad \text{normal vector } \vec{n} = \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|} = \frac{(-h_u, -h_v, 1)}{\sqrt{(1+h_u^2+h_v^2)}}$$

with the shape operator:

$$P = I^{-1} II = \begin{bmatrix} E & F \\ F & G \end{bmatrix}^{-1} \begin{bmatrix} L & M \\ M & N \end{bmatrix} = \begin{bmatrix} h_{uu}(1+h_v^2) - h_{uv}h_vh_u & h_{uv}(1+h_v^2) - h_{vv}h_vh_u \\ h_{uv}(1+h_u^2) - h_{uu}h_uh_v & h_{vv}(1+h_u^2) - h_{uv}h_uh_v \end{bmatrix} \frac{1}{(1+h_u^2+h_v^2)^{3/2}}$$

Note that $\vec{r} = (u, v, h[u, v])$ defines a Monge patch.

and Gaussian and Mean Curvatures are:

$$K = \frac{LN - M^2}{EG - F^2} \quad \text{and} \quad H = \frac{GL - 2FM - EN}{2(EG - F^2)}$$

Principal Curvatures $\kappa_1 \kappa_2$ along the principal directions forming the lines of curvature for a function $h(u, v)$ in Cartesian coordinates are the eigenvalues of P , and for the Monge patch are:

$$\kappa_1 = \frac{h_{vv} + h_{vv}h_u^2 - 2h_vh_uh_{uv} + h_{uu} + h_v^2h_{uu}}{2(1+h_v^2+h_u^2)^{\frac{3}{2}}} - \sqrt{\frac{(h_{vv} + h_{vv}h_u^2 - 2h_vh_uh_{uv} + h_{uu} + h_v^2h_{uu})^2}{4(1+h_v^2+h_u^2)^3} + \frac{h_{uv}^2 - h_{vv}h_{uu}}{(1+h_v^2+h_u^2)^2}}$$

$$\kappa_2 = \frac{h_{vv} + h_{vv} h_u^2 - 2h_v h_u h_{uv} + h_{uu} + h_v^2 h_{uu}}{2(1 + h_v^2 + h_u^2)^{\frac{3}{2}}} + \sqrt{\frac{(h_{vv} + h_{vv} h_u^2 - 2h_v h_u h_{uv} + h_{uu} + h_v^2 h_{uu})^2}{4(1 + h_v^2 + h_u^2)^3} + \frac{h_{uv}^2 - h_{vv} h_{uu}}{(1 + h_v^2 + h_u^2)^2}}$$

Note that $\kappa_1 \kappa_2 > 0$ if the Hessian is positive definite: $h_{uu} h_{vv} - h_{uv}^2 > 0$

Monge/Local Astigmatism $\kappa_1 - \kappa_2$

$$A = 2 \sqrt{\frac{(h_{vv} + h_{vv} h_u^2 - 2h_v h_u h_{uv} + h_{uu} + h_v^2 h_{uu})^2}{4(1 + h_v^2 + h_u^2)^3} + \frac{h_{uv}^2 - h_{vv} h_{uu}}{(1 + h_v^2 + h_u^2)^2}}$$

Mean curvature $(\kappa_1 + \kappa_2)/2$

$$H = \frac{h_{vv} + h_{vv} h_u^2 - 2h_v h_u h_{uv} + h_{uu} + h_v^2 h_{uu}}{2(1 + h_v^2 + h_u^2)^{\frac{3}{2}}}$$

Gaussian Curvature $K = \kappa_1 \kappa_2$

$$K = \frac{h_{vv} h_{uu} - h_{uv}^2}{(1 + h_v^2 + h_u^2)^2}$$

The principal curvatures are also expressible as

$$\kappa_1, \kappa_2 = H \pm \sqrt{H^2 - K} = H \pm A/2$$

In polar coordinates this gets more cumbersome, though the ubiquitous denominator of first derivatives (the metric) is quite simple, so for $r > 0$:

$$h_u = h_r \cos(t) - \sin(t) \frac{h_t}{r}$$

$$h_v = h_r \sin(t) + \cos(t) \frac{h_t}{r}$$

$$(1 + h_v^2 + h_u^2) = \left(1 + h_r^2 + \left(\frac{h_t}{r}\right)^2\right)$$

The conversion for the second derivatives (Hessian) is

$$h_{uu} = h_{rr} - \sin^2(t) \left(h_{rr} - \frac{h_r}{r} - \frac{h_{tt}}{r^2} \right) + 2 \cos(t) \sin(t) \left(\frac{h_t}{r^2} - \frac{h_{rt}}{r} \right)$$

$$h_{vv} = h_{rr} - \cos^2(t) \left(h_{rr} - \frac{h_r}{r} - \frac{h_{tt}}{r^2} \right) - 2 \cos(t) \sin(t) \left(\frac{h_t}{r^2} - \frac{h_{rt}}{r} \right)$$

$$h_{uv} = \cos(t) \sin(t) \left(h_{rr} - \frac{h_r}{r} - \frac{h_{tt}}{r^2} \right) + (\sin^2(t) - \cos^2(t)) \left(\frac{h_t}{r^2} - \frac{h_{rt}}{r} \right)$$

Simplified formulae for surfaces of revolution (axisymmetric surfaces with elevation $z(r)$ where r is the radius from the axis of revolution) and all derivatives in θ are zero:

$$z_\theta = 0, \quad z_{r\theta} = 0, \quad \text{and} \quad z_{\theta\theta} = 0$$

"Instantaneous or Tangential" curvature, polar coordinates:

$$\kappa_1 = \frac{z_{rr}}{(1 + z_r^2)^{3/2}}$$

which actually the formula for the first principal curvature for a surface of revolution (axisymmetric) parameterized by $z=z(r)$ The other principal curvature in that case is actually the

"Axial or Sagittal" curvature ", polar coordinates:

$$\kappa_2 = \frac{z_r}{r \sqrt{1 + z_r^2}}, \text{ with } z_r \rightarrow 0 \text{ at } r \rightarrow 0 \text{ implies } \kappa_2 \rightarrow z_{rr}$$

Note also that the tangential curvature is the derivative of the sagittal curvature in this case.

Consider a simple paraboloid, $z(r)=r^2$. The ratio between $\kappa_1/\kappa_2 = (1+4r^2)$, showing that for $r \ll 1$, the two are similar, but as the radius increases, the axial curvature becomes larger than the tangential curvature. Since the two principal curvatures are orthogonal, that also allows us to realize that the axial curvature is in the radial direction, whereas the tangential curvature is in the circumferential direction.

Mean Power is the arithmetic average $(\kappa_1 + \kappa_2)/2$. H is traditionally used.

$$H = \frac{1}{2} \left[\frac{z_r}{r \sqrt{1+z_r^2}} + \frac{z_{rr}}{(1+z_r^2)^{3/2}} \right]$$

Gaussian Power is the product $\kappa_1 \kappa_2$ of the principal curvatures. It's unfortunate that it isn't the geometric mean instead of its square...

$$K = \frac{z_r z_{rr}}{r(1+z_r^2)}$$

So in order to get a comparable "power" in diopters, the square root of K is used in our representations.

So called Monge Astigmatism $\kappa_1 - \kappa_2$:

$$A = \left[\frac{z_r}{r \sqrt{1+z_r^2}} - \frac{z_{rr}}{(1+z_r^2)^{3/2}} \right]$$

Warp/Oblique power, same as axial but with z_θ/r instead of z_r
This last definition is non-standard and defined here as:

$$K = \frac{z_\theta}{r^2 \sqrt{1+(z_\theta/r)^2}},$$

The Cartesian equation for the lines of curvature is given as the solutions of the ODE:

$$a = EM - FL, \quad b = EN - GL, \quad c = FN - GM$$

where the equation is given by

$$\begin{vmatrix} dv^2 & -dv du & du^2 \\ E & F & G \\ L & M & N \end{vmatrix} = 0 \quad \text{or} \quad c dv^2 + b dv du + a du^2 = 0 \quad \text{or}$$

$$\frac{v'(t)^2 \left(h_v h_{vv} h_u - h_{uv} - h_v^2 h_{uv} \right)}{\sqrt{1 + h_v^2 + h_u^2}} + \frac{u'(t) v'(t) \left(h_{vv} + h_{vv} h_u^2 - h_{uu} - h_v^2 h_{uu} \right)}{\sqrt{1 + h_v^2 + h_u^2}} + \frac{u'(t)^2 \left(h_{uv} + h_u^2 h_{uv} - h_v h_u h_{uu} \right)}{\sqrt{1 + h_v^2 + h_u^2}} = 0$$

with $h_u = \frac{\partial h}{\partial u}$, $h_v = \frac{\partial h}{\partial v}$ which has two solution sets as the solution to $u'(t) = m(u(t), v(t)) v'(t)$ where $u(t)$, $v(t)$ describe parametric curves in t using the two roots of the quadratic solution with some rearrangement $m(u, v)$:

$$m(u, v) = (-b - \text{sign}(b) \sqrt{(b^2 - 4ac)}) / 2a$$

or

$$m(u, v) = 2c / (-b - \text{sign}(b) \sqrt{(b^2 - 4ac)})$$

with

$$a = \left(\frac{(1 + h_u^2) h_{uv}}{\sqrt{1 + h_v^2 + h_u^2}} - \frac{h_v h_u h_{uu}}{\sqrt{1 + h_v^2 + h_u^2}} \right)$$

$$c = \left(\frac{h_v h_u h_{vv}}{\sqrt{1 + h_v^2 + h_u^2}} - \frac{(1 + h_v^2) h_{uv}}{\sqrt{1 + h_v^2 + h_u^2}} \right)$$

$$b = \left(\frac{h_{vv} (1 + h_u^2)}{\sqrt{1 + h_v^2 + h_u^2}} - \frac{(1 + h_v^2) h_{uu}}{\sqrt{1 + h_v^2 + h_u^2}} \right)$$

Principal Directions

The "lines of curvature" describing the directions of the principal curvatures are the integral lines of the vector field of the principal directions. It is much easier computationally to get the principal directions and plot the vector field, avoiding solving the ODE. The lines of curvature in the axisymmetric case are given by the meridians and circles of longitude. The principal directions can be plotted as a vector field in (u,v) . Using the following formula:

$$\kappa = \frac{v^2 h_{vv} + 2vuh_{uv} + u^2 h_{uu}}{(v^2(1+h_v^2) + 2vuh_v h_u + u^2(1+h_u^2))\sqrt{(1+h_v^2+h_u^2)}}$$

where κ is the curvature at any point (u,v) and the angle ϕ describes the direction of the curvature with relation to the principal curvature using Euler's equation. $\kappa = \kappa_1 \cos^2(\phi) + \kappa_2 \sin^2(\phi)$ Then we have

$$\tan \phi = \sqrt{\frac{\kappa_2 - \kappa}{\kappa_1 - \kappa}} \quad \text{which becomes indeterminate at umbilical points because}$$

there $\kappa_1 = \kappa_2$ but $\kappa_1 > \kappa > \kappa_2$, so $\kappa = \kappa_1 = \kappa_2$

$$\tilde{R} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \text{sign}(u)\sqrt{\frac{\kappa_1 - \kappa}{\kappa_1 - \kappa_2}} & \text{sign}(v)\sqrt{\frac{\kappa - \kappa_2}{\kappa_1 - \kappa_2}} & 0 \\ -\text{sign}(v)\sqrt{\frac{\kappa - \kappa_2}{\kappa_1 - \kappa_2}} & \text{sign}(u)\sqrt{\frac{\kappa_1 - \kappa}{\kappa_1 - \kappa_2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{defines the rotation}$$

in ϕ

We can plot the vector field for one principal direction as (u, v, u_t, v_t) with $(u_t, v_t) = \kappa_1 \cdot \tilde{R} \cdot (u/r, v/r, 0)$

and the other at 90 degrees rotation, or with $(u_t, v_t) = \kappa_2 \cdot \tilde{R}_0 \cdot \tilde{R} \cdot (u/r, v/r, 0)$

$$\tilde{R}_0 = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{defines a 90 degree rotation}$$

Above we have pretended that ϕ is an angle in the $(u,v,0)$ plane, but of course it is not.

These calculation should be made in the tangent plane of the surface, not the horizontal $(u, v, 0)$ plane. The coordinate system of $\{u, v, h(u, v)\}$ has basis e_1, e_2, e_3 . The tangent plane is orthogonal to the normal vector $\vec{n} = \frac{(-h_u, -h_v, 1)}{\sqrt{(1+h_u^2+h_v^2)}}$ we can then take the vector in the horizontal $(u/r, v/r, 0)$ plane and subtract off the tilt by defining $\tilde{M} = \tilde{I} - \vec{n} \otimes \vec{n}$ with the identity matrix and the outer product, giving:

$$\vec{r} = \tilde{M} \cdot \left(\frac{u}{\sqrt{u^2+v^2}}, \frac{v}{\sqrt{u^2+v^2}}, 0 \right) \text{ and then rotate it in the tangent plane with}$$

We also need the metric tensor for the coordinate system of the tangent plane. We define/construct a new basis:

$$\vec{e}'_1 = \frac{\vec{r} - (\vec{n} \cdot \vec{r}) \vec{n}}{\|\vec{r} - (\vec{n} \cdot \vec{r}) \vec{n}\|} \quad \vec{e}'_2 = \vec{n} \times \vec{e}'_1 \quad \vec{e}'_3 = \vec{n} \text{ with the resulting matrix}$$

$$\tilde{J} = \begin{bmatrix} \vec{e}_1 \cdot \vec{e}'_1 & \vec{e}_1 \cdot \vec{e}'_2 & \vec{e}_1 \cdot \vec{e}'_3 \\ \vec{e}_2 \cdot \vec{e}'_1 & \vec{e}_2 \cdot \vec{e}'_2 & \vec{e}_2 \cdot \vec{e}'_3 \\ \vec{e}_3 \cdot \vec{e}'_1 & \vec{e}_3 \cdot \vec{e}'_2 & \vec{e}_3 \cdot \vec{e}'_3 \end{bmatrix} \text{ with } \begin{matrix} \vec{e}_1 = (1, 0, 0) \\ \vec{e}_2 = (0, 1, 0) \\ \vec{e}_3 = (0, 0, 1) \end{matrix} \text{ for the original coordinates.}$$

Finally we get

$$\vec{r}'_1 = [\tilde{J}^{-1}]^T \tilde{R} [\tilde{J}]^T \cdot \vec{r} \text{ where we take the first two components for the vector field for the first principal direction in } (u, v), \text{ or } (u, v, \kappa_1 \vec{r}'_1(1), \kappa_1 \vec{r}'_1(2))$$

For the other direction, we rotate the matrix R in the u, v plane, so we get $\vec{r}'_2 = [\tilde{J}^{-1}]^T \tilde{R}_0 \tilde{R} [\tilde{J}]^T \cdot \vec{r}$ whose first two components

for the vector field for the second principal direction in (u, v) , or $(u, v, \kappa_2 \vec{r}'_2(1), \kappa_2 \vec{r}'_2(2))$

Note that as a sanity check for calculations, we should have:

$$(\tilde{M} \cdot \vec{r}'_1) \cdot (\tilde{M} \cdot \vec{r}) - (\tilde{M} \cdot \vec{r}) \cdot (\tilde{M} \cdot \vec{r}) * \text{sign}(u) \sqrt{\frac{\kappa_1 - \kappa_2}{\kappa_1 - \kappa_2}} \rightarrow 0 \text{ and } \vec{n} \cdot (\vec{r}'_1 - \vec{r}) \rightarrow 0$$

$$(\tilde{M} \cdot \vec{r}'_2) \cdot (\tilde{M} \cdot \vec{r}) + (\tilde{M} \cdot \vec{r}) \cdot (\tilde{M} \cdot \vec{r}) * \text{sign}(v) \sqrt{\frac{\kappa - \kappa_2}{\kappa_1 - \kappa_2}} \rightarrow 0 \text{ and } \vec{r}'_1 \cdot \vec{r}'_2 - \vec{r} \cdot \vec{r} \rightarrow 0$$

The “Axial Power or Axial Distance”, as it has been styled, is not defined for off center or non-axisymmetric case, but there are formal alternatives that have been proposed, for example, defining the following, based on the surface vectors and normals:

$\vec{v}=(u,v,h(u,v))$ with normal $\vec{n}=\frac{(-h_u,-h_v,1)}{\sqrt{(1+h_u^2+h_v^2)}}$ and at the origin

the values $\vec{v}_0=\vec{v}(0,0)$ and $\vec{n}_0=\vec{n}(0,0)$ then we can define the vectors

$\vec{ac}=\vec{n}_0\times(\vec{v}-\vec{v}_0)$ and $\vec{an}=\vec{n}\times\vec{n}_0$ which lead to one of three “radii”

$$\text{Definition 1}=\sqrt{\frac{\vec{ac}\cdot\vec{ac}}{\vec{an}\cdot\vec{an}}}=\frac{|\vec{ac}|}{|\vec{an}|}$$

$$\text{Definition 2}=\frac{\vec{ac}\cdot\vec{an}}{\sqrt{\vec{an}\cdot\vec{an}}}=\frac{\vec{ac}\cdot\vec{an}}{|\vec{an}|}$$

$$\text{Definition 3}=\sqrt{\frac{\vec{ac}\cdot\vec{ac}}{(\vec{n}_0\times\vec{X})\cdot(\vec{n}_0\times\vec{X})}} \text{ with } \vec{X}=\frac{(\vec{ac}\times\vec{n})\times\vec{ac}}{\|(\vec{ac}\times\vec{n})\times\vec{ac}\|}$$

or

$$\text{Definition 3}=\frac{\|((\vec{ac}\times\vec{n})\times\vec{ac})\| |\vec{ac}|}{\|(\vec{n}_0\times((\vec{ac}\times\vec{n})\times\vec{ac}))\|}$$

(Klein SA, Mandell RB, Barsky BA. Representing corneal shape.)

Picking Definition 1, with $\vec{n}_0=(0,0,1)$ which corresponds to no axial angular misalignment, only (u,v) misalignment, leads to

$$\kappa = \left(\frac{\sqrt{(h_u^2 + h_v^2)}}{\sqrt{((v - v_0)^2 + (u - u_0)^2)(1 + h_v^2 + h_u^2)}} \right)$$

which is very similar to the polar coordinate axisymmetric version

given earlier: $\kappa = \frac{h_r}{r\sqrt{1+h_r^2}}$

Note that with $\vec{n}_0=(0,0,1)$, that the tangent plane of the surface is orthogonal to the VK axis as well as centered. Definition 1 explicitly assumes that rays stay in the meridional plane.

Picking Definition 2, with $\vec{n}_0=(0,0,1)$ which also corresponds to no axial angular misalignment, only (u,v) misalignment, but with a different geometric interpretation, leads to

$$\kappa = \left(\frac{\sqrt{(h_v^2 + h_u^2)}}{(h_v(v - v_0) + h_u(u - u_0))\sqrt{(1 + h_v^2 + h_u^2)}} \right)$$

while Definition 3 (still with $\vec{n}_0=(0,0,1)$) is quite complicated.

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